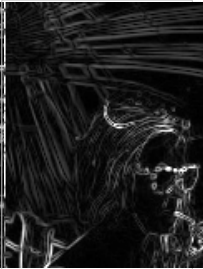
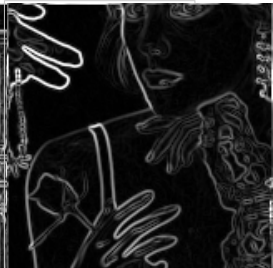
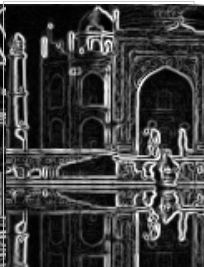
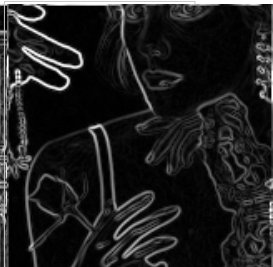


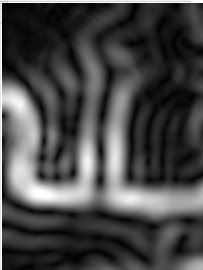
Section 9 Scenario A natural images

BSDS500 real-image comparison with validation-tuned classical baselines

Five BSDS500 test crops. Clean qualitative panels. Metrics over SNR levels clean, 20 dB, 10 dB, and 5 dB.

Gradient magnitude, clean real images

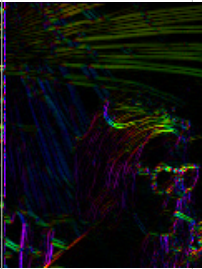
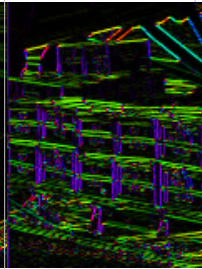
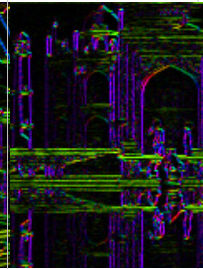
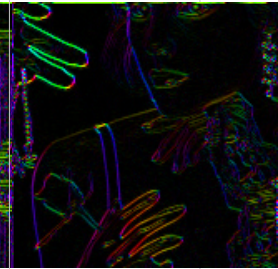
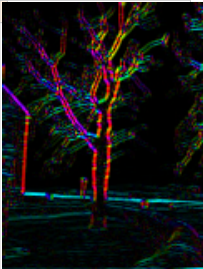
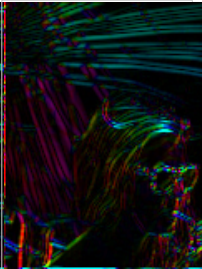
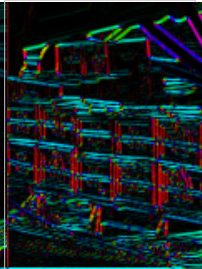
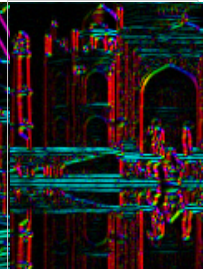
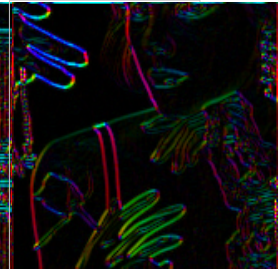
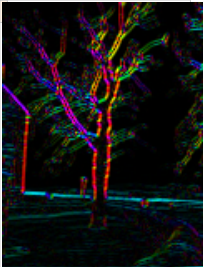
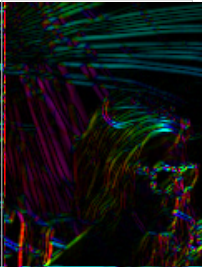
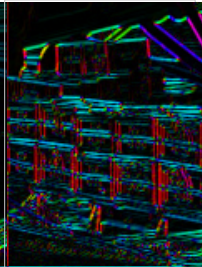
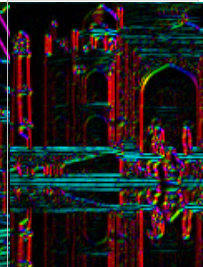
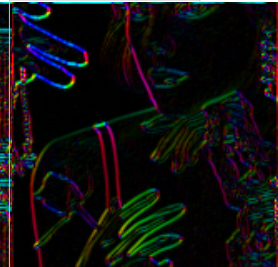
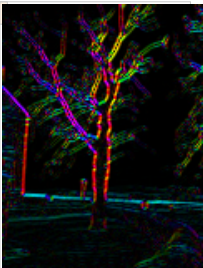
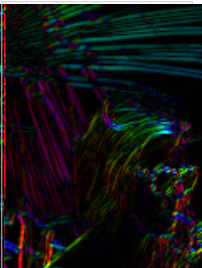
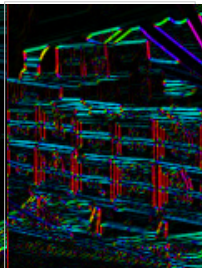
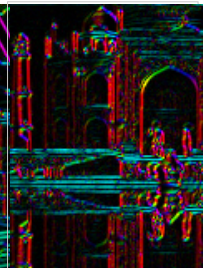
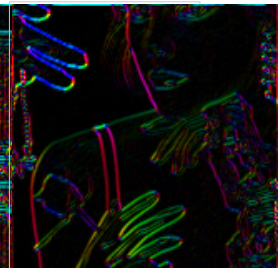
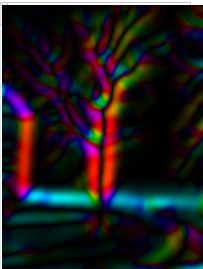
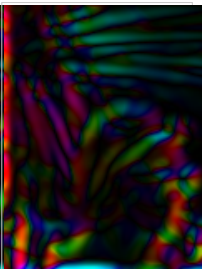
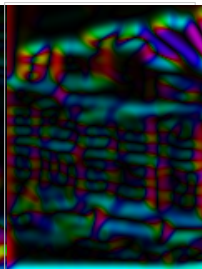
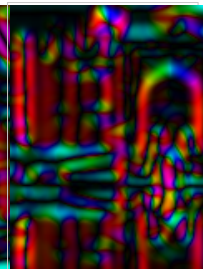
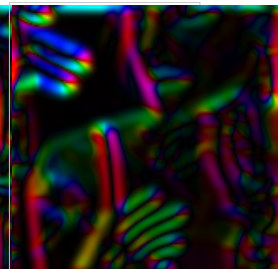
Method	92014-1	81090-2	188025-3	288024-4	198087-5
					
Roberts					
Prewitt					
Sobel					
Scharr					

DoG					
Farid-Simoncelli					
Square SG					
WVF					

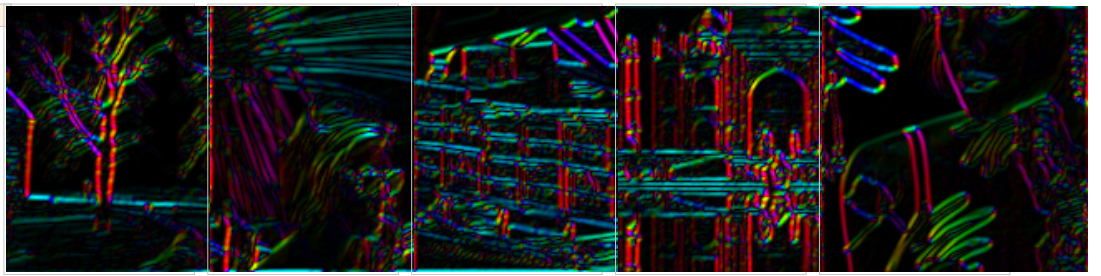
Section 9 Scenario A natural images

Orientation maps. Hue encodes edge angle modulo π and brightness follows normalized magnitude.

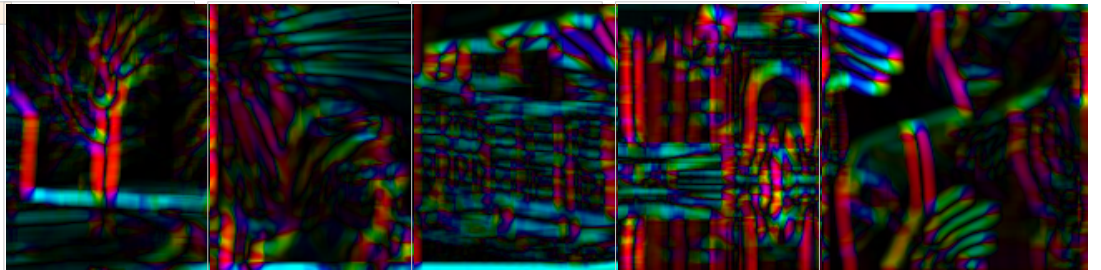
Gradient orientation, clean real images

Method	92014-1	81090-2	188025-3	288024-4	198087-5
Roberts					
Prewitt					
Sobel					
Scharr					
DoG					

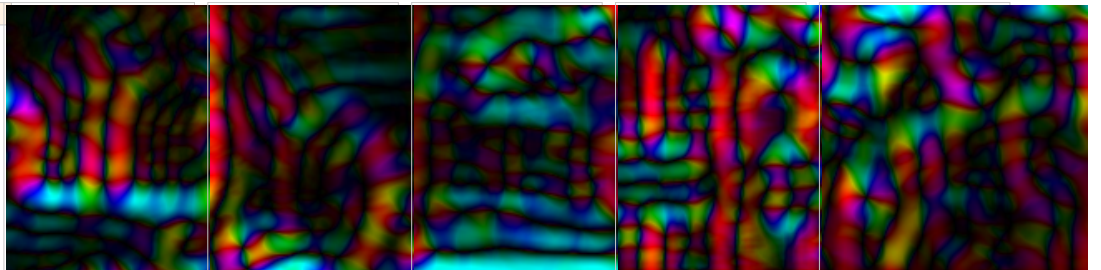
Farid-Simoncelli



Square SG

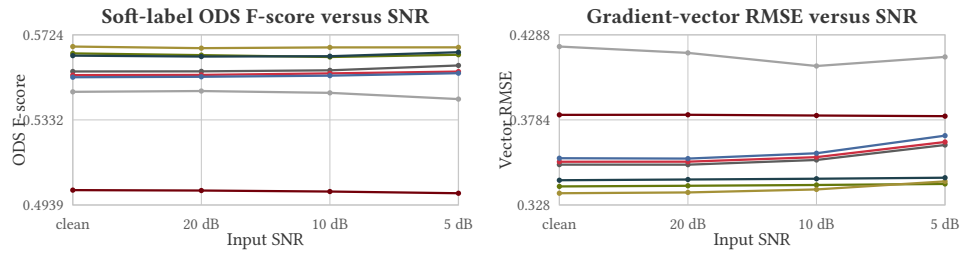


WVF



Section 9 Scenario A natural images

Quantitative summary across 5 crops. Noisy metrics average over 10 AWGN draws per crop.

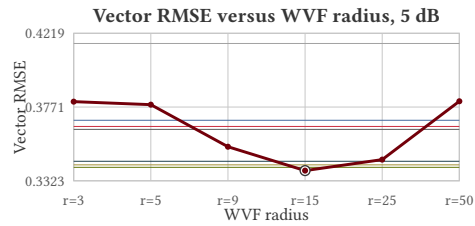
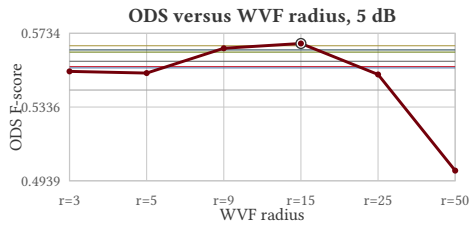
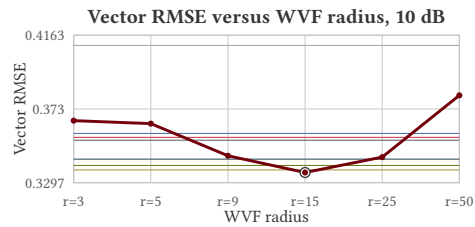
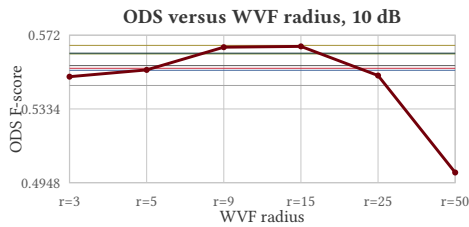
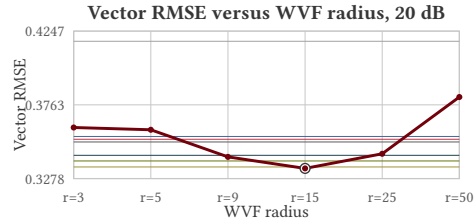
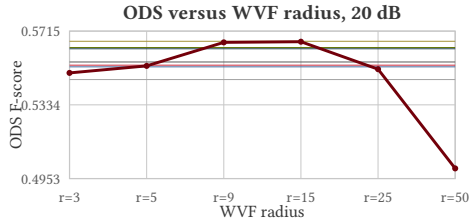
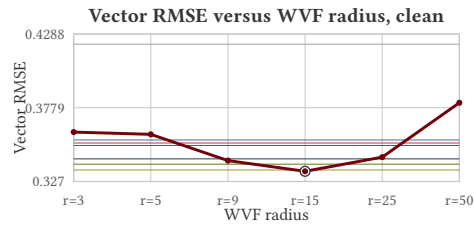
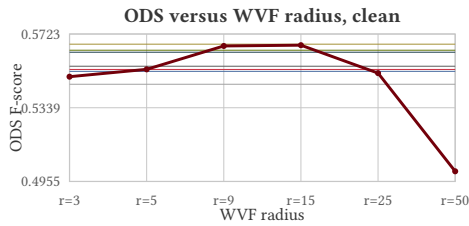


Metric	Roberts	Prewitt	Sobel	Scharr	DoG	Farid-Simoncelli	Square SG	WVF
ODS clean	0.5461	0.5555	0.5539	0.5529	0.5639	0.567	0.5628	0.5008
ODS 10 dB	0.5457	0.5561	0.5547	0.5536	0.5622	0.5666	0.5626	0.5001
RMSE clean	0.4218	0.3519	0.3536	0.3557	0.339	0.3349	0.3426	0.3814
RMSE 10 dB	0.4103	0.3546	0.3563	0.3586	0.3398	0.3372	0.3435	0.381

Section 9 Scenario A natural images

WVF radius-trace follow-up on the fixed BSDS crop set.

Garnet trace points sweep WVF across $(r, d) = (3, 5), (5, 9), (9, 11), (15, 11), (25, 11), (50, 11)$. Thin horizontal lines are the fixed baseline methods. Outlined WVF points beat the best fixed baseline at that metric and SNR.

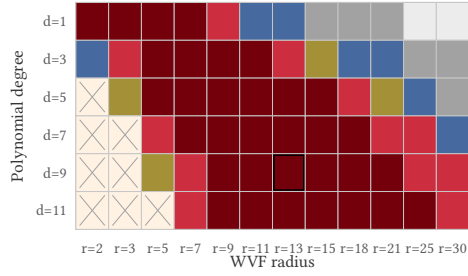


Section 9 Scenario A natural images

Fine-grained WVF radius and degree sweep on the fixed BSDS crop set.

The heatmap shows 10 dB gradient-vector RMSE across the feasible (r, d) grid. Crossed cells were skipped by the Section 7 conditioning gate. The outlined cell is the annotated optimum.

10 dB gradient-vector RMSE across the feasible WVF grid



Annotated optimum. $r=13$, $d=9$ with RMSE 0.335645.

Driver assessment. the optimum BSDS radius stays in a narrow band from $r=11$ to $r=13$ across the SNR sweep, so feature scale is the dominant constraint and the noise floor only perturbs the optimum by one grid step.